

Phase 2: time trial, timing and results

BeamMP Race Manager Client: dhardesty99

The core of the whole product. Records, XP, teams and challenges all sit on top of what gets built here, so anything wrong in this phase is wrong in every phase after it.

1 What gets delivered

From his document, unchanged:

- Time Trial, with a lap count if he wants laps
- Controller or Wheel selected before the run
- Teleport to the start, and the run arms when he drives through the trigger
- End Race stops all timing at once
- A split recorded at every checkpoint
- Missed checkpoints caught and penalised
- The full results screen
- No live leaderboard during the run

Plus two things he asked for after the plan was written:

- **Penalties only apply during a race.** Free driving costs nothing. That decides the shape of the race state machine, so it belongs here rather than in phase 3.
- **His colour scheme.** Dark red, black and grey, with a little orange and yellow for contrast. The results screen is the first thing worth styling.

2 The one rule this phase turns on

The server clock is the only clock. Not one timing value comes from the client. The client reports that it crossed a gate; the server decides when.

That is not caution for its own sake. The client mod ships to every player as a readable zip, so a lap time the client calculates is a lap time the client can choose.

The cost is the network round trip. A crossing stamped on arrival gives a player on 200 ms ping 200 ms of lag in the stamp. Two ways to deal with it, and the second is better:

1. Stamp on arrival. Simple, and it quietly penalises high ping.
2. The client sends its own frame timestamp alongside the crossing, the server keeps a rolling estimate of that player's offset, and corrects. The client can still lie, but only inside a window the server already believes, and a lie that drifts gets caught.

Going with the second, with the window clamped. Worth writing down now because it is the single decision the rest of the phase hangs on.

3 Race state machine

One race state per player for time trial. Phase 4 turns this into a shared grid, so the shape is chosen now to survive that.

idle → armed → running → finished, or running → abandoned

From	To	On
idle	armed	picks a course, teleports to the start
armed	running	drives through the start gate
running	running	crosses a gate, split recorded
running	finished	crosses the finish on the final lap
running	abandoned	End Race, disconnect, or leaves the level
finished	idle	closes the results

Penalties only exist in running. That is his change, and it falls out of the machine rather than being bolted onto it.

4 Splits and laps

Stored on the server, per run: the course and mode, the lap count and current lap, start and finish stamps, a split per gate per lap, the missed gates per lap, and the penalties with their reasons.

Splits are seconds from the start of the run, not deltas. Deltas are a presentation choice and can always be computed; a stored delta cannot be recovered into an absolute once a gate is missed.

5 Missed checkpoints

A gate is missed when a later gate fires while an earlier one has no split for this lap. That catches both a deliberate cut and a car tunnelling a gate at speed.

The penalty is per missed gate, and the reason carries the gate number, so the breakdown in the results reads as something a person can argue with.

Circuit courses wrap, so gate 1 firing after gate 29 is a lap, not a miss. The lap boundary is the start gate, which is why the start line is stored separately from the checkpoint list.

6 Results screen

Only after everyone has finished. No live leaderboard, which was his call and removes the largest network cost in the system.

Panel	Notes
Corrected time	penalties included. This decides the result.
Click a name	that player's penalties, each with its reason
Dirty time	clean time, information only, decides nothing
Overall best lap	
Personal best lap	click to expand every lap

Panel	Notes
Diff to the car ahead	
Diff to the leader	
Personal best split	click to expand every split

7 Protocol additions

Direction	Channel	Carries
client to server	<code>race.arm</code>	course, mode, lap count
client to server	<code>race.end</code>	nothing
client to server	<code>cp.hit</code>	gains a client frame time, for offset correction
server to client	<code>race.state</code>	the state machine, on change only
server to client	<code>race.split</code>	the split just recorded, so the HUD can show it
server to client	<code>race.result</code>	the whole results payload, once
server to client	<code>race.teleport</code>	where to put the car when arming

8 Performance

Same rules as phase 1, and one new risk. A split is one small message per gate crossing. Twenty-nine gates against a full grid is phase 4's problem, but the message shape is decided here, so it is sized for that now: a split is an index and a number, not a record.

Nothing is recalculated on a timer. A split is computed when a gate fires and never again. The results payload is built once, when the last driver finishes.

Measured against the phase 1 baseline: **128 average, 108 one per cent low**, taken on his server.

9 Testing

The mock host gets a clock that can be driven by hand, so a run can be played through in a test without waiting for real seconds. Cases worth having before the code is written:

- A clean lap. Splits in order, corrected equals dirty.
- A missed gate. Penalty logged against the right gate number.
- A gate crossed twice. The second crossing ignored.
- Gates crossed out of order.
- Disconnect mid run. Run abandoned, nothing half written to disk.
- End Race mid run. All timing stops at once.
- A client reporting a gate it cannot have reached yet.
- A client reporting a frame time far outside the believed window.
- Two runs on one course by one player. Personal best updates only when it should.

Then on his server, with two drivers, which is the part that is still blocked.

10 What is needed from him

Item	Needed for	Status
Second account	two driver testing	agreed, never received. Blocks sign off.
Captured routes	something real to time	tool is his, waiting on him
Race classes	phase 5, shapes the record now	waiting
Logo files	results screen styling	sent, colours noted

11 Order of work

1. Race state machine, server side, with tests
2. Clock and offset correction
3. Splits and lap counting
4. Missed checkpoint detection
5. Arming, teleport, and the End Race button
6. Results payload
7. Results screen, in his colours
8. FPS measured against the baseline

The CoPilot camera gets a quiet look during this phase. It is the least known part of the whole job, and week eleven is the wrong time to find out it is hard.